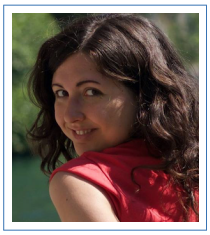
ALICE RAFFAELE

November 09, 2026

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Personal profile



Current position

Postdoctoral researcher at the Department of Information Engineering of the University of Padova, Via Giovanni Gradenigo 6/b, 35135 Padova, Italy.

Research interests: Mathematical programming, combinatorial optimization, routing problems under uncertainty, exact algorithms, meta-heuristics, operations research in education, enumeration problems.

Indexed research and bibliometric indicators:

- ORCID: `https://orcid.org/0000-0003-1323-7544`
- Scopus: h-index: 5; total number of citations: 61; since 2021: 61; without self: 50.
  Profile: `https://www.scopus.com/authid/detail.uri?authorId=57486817000`
- Web of Science: h-index: 4; total number of citations: 40; since 2021: 40.
  Profile: `http://www.webofscience.com/wos/author/record/ABY-6928-2022`
- Google Scholar: h-index: 5; total number of citations: 109; since 2021: 106.
  Profile: `https://scholar.google.com/citations?user=v5U6Aq0AAAAJ&hl=it&oi=ao`

Education

Jun, 2018 – Oct, 2018 **24 CFU Teacher Training**, *University of Verona*, Italy.
7 modules in the areas of Pedagogy, Special pedagogy and Didactics of inclusion, Psychology, Anthropology, Teaching methodologies and technologies.

Oct, 2013 – Mar, 2016 **Master of Science, Computer Science and Engineering**, *University of Brescia*, Italy, 110/110 cum laude, March 16, 2016.
Thesis: "Time constrained Vehicle Routing Problem"
Advisor: Prof. Renata Mansini.
Optimization classes attended: *Optimization algorithms* by Prof. Renata Mansini.

Sep, 2010 – Oct, 2013 **Bachelor of Science, Computer Science and Engineering**, *University of Brescia*, Italy, 108/110, October 30, 2013.
Thesis: "La cifratura omomorfica" (i.e., "The homomorphic encryption")
Advisor: Prof. Luca Giuzzi.
Optimization classes attended: *Operations research* by Prof. Renata Mansini.

Sep, 2005 – Jun, 2010 **High School Scientific Diploma**, *Liceo Scientifico Statale G. Galilei*, Palazzolo sull'Oglio (BS), Italy, 92/100, June, 2010.
Thesis: "Alla ricerca di una nuova identità — La figura della donna nella letteratura dalla Rivoluzione Francese in poi" (i.e., "Searching for a new identity – The role of women in literature, from the French Revolution and on").

Research

Research appointments

Apr, 2025 – present **Postdoctoral researcher in Operations Research**, *University of Padova*, Italy.
Mathematical Programming, Combinatorial Optimization, Routing problems under uncertainty.
Advisor: Prof. Domenico Salvagnin

Apr, 2023 – Mar, 2025 **Postdoctoral researcher in Operations Research**, *University of Calabria*, Italy.
Mathematical Programming, Combinatorial Optimization, Routing problems under uncertainty.
Advisors: Prof. Demetrio Laganà and Prof. Roberto Musmanno

Apr, 2022 – Mar, 2022 **Postdoctoral researcher in Operations Research**, *University of Verona*, Italy.
Mathematical Programming, Combinatorial Optimization, Algorithms, Temporal networks.
Advisor: Prof. Romeo Rizzi

Apr, 2021 – Mar, 2022 **Research fellow in Operations Research**, *University of Verona*, Italy.
Mathematical Programming, Combinatorial Optimization, Algorithms, Temporal networks.
Advisor: Prof. Romeo Rizzi

Nov, 2017 – Mar, 2022 **Ph.D. in Mathematics (Operations Research)**, *University of Trento and University of Verona*, Italy.
Combinatorial Optimization, Mathematical Programming, Algorithms.
Thesis: "From optimization to listing: theoretical advances in some enumeration problems".
Advisor: Prof. Romeo Rizzi
Defense date: March 30, 2022.

Visitings

Apr, 2023 – Mar, 2025 **Visiting researcher**, *University of Padova*, Padova, Italy.
Mathematical Programming, Combinatorial Optimization, Routing problems under uncertainty.
Advisor: Prof. Roberto Roberti

Oct, 2019 – Dec, 2019 **Visiting researcher**, *National Institute of Informatics*, Tokyo, Japan.
Enumeration algorithms and listing problems.
Advisor: Prof. Takeaki Uno

May, 2019 – June, 2019 **Visiting researcher**, *Université Paris-Est Marne-la-Vallée*, Paris, France.
Bioinformatics and string algorithms (problems related to permutation patterns and graphs). Advisors: Prof. Stéphane Vialette and Prof. Laurent Bulteau

Sep, 2015 – Jan, 2016 **Master's thesis intern**, *Université Laval*, Québec City, Canada.
Vehicle routing problem and metaheuristics. Advisors: Prof. Jean-François Côté and Prof. Renata Mansini

Ongoing research projects

Sep, 2025 – present **Proyectos de Generación de Conocimiento 2024**, *University of Seville, Spain*.
Title: RE-POWERED – Renewable Energy Planning and Optimization for Wind Energy REcycling and Decommissioning.
Role: Participant.

Publications

Journal articles

2026 **Alice Raffaele**, Demetrio Laganà, and Roberto Roberti. Robust policies for a multi-period fleet sizing problem with demand uncertainty. *European Journal of Operational Research*, volume 332 (2), pages 657–675. Elsevier, 2026.

2025 **Alice Raffaele** and Romeo Rizzi. New theoretical results on the Monotone Boolean Duality and the Monotone Boolean Dualization problems. *Discrete Applied Mathematics*, volume 361, pages 347–369. Elsevier, 2025.

2025 Gabriella Colajanni, Alessandro Gobbi, Marinella Picchi, **Alice Raffaele**, and Eugenia Taranto. An Operations Research–Based Teaching Unit for Grade 12: The ROAR Experience, Part III. *INFORMS Transactions on Education*, volume 26, pages 12–33. INFORMS, 2025.

2024 **Alice Raffaele**, Romeo Rizzi, and Takeaki Uno. Listing the bonds of a graph in $\widetilde{O}(n)$–delay. *Discrete Applied Mathematics*, volume 348, pages 105–121. Elsevier, 2024.

2024 Eugenia Taranto, Gabriella Colajanni, Alessandro Gobbi, Marinella Picchi, and **Alice Raffaele**. Fostering students' modelling and problem-solving skills through operations research, digital technologies and collaborative learning. *International Journal of Mathematical Education in Science and Technology*, volume 55, pages 1957–1998. Taylor & Francis, 2024.

2024 Jean-François Côté, Renata Mansini, and **Alice Raffaele**. Multi-period time window assignment for attended home delivery. *European Journal of Operational Research*, volume 316, pages 295–309. Elsevier, 2024.

2024 Gabriella Colajanni, Alessandro Gobbi, Marinella Picchi, **Alice Raffaele**, and Eugenia Taranto. An Operations Research–based Teaching unit for Grade 11: The ROAR Experience, Part II. *INFORMS Transactions on Education*, volume 24, pages 238–258. INFORMS, 2024.

2023 Matteo Zavatteri, **Alice Raffaele**, Dario Ostuni, and Romeo Rizzi. An interdisciplinary experimental evaluation on the disjunctive temporal problem. *Constraints*, volume 28, pages 1–12. Springer, 2023.

2023 **Alice Raffaele** and Matteo Zavatteri. Reducing the number of disjuncts in DTPs. *Information and Computation*, volume 293, page 105048. Elsevier, 2023.

2023 Gabriella Colajanni, Alessandro Gobbi, Marinella Picchi, **Alice Raffaele**, and Eugenia Taranto. An Operations Research–based Teaching Unit for Grade 10: The ROAR Experience, Part I. *INFORMS Transactions on Education*, volume 23, pages 104–120. INFORMS, 2023.

2021 **Alice Raffaele** and Alessandro Gobbi. Teaching operations research before university: A focus on grades 9–12. In *Operations Research Forum*, volume 2, page 13. Springer, 2021.

2021 **Alice Raffaele**. Becoming Visible: Why We Should be Better Communicators Now. *SN Operations Research Forum*, volume 1, 7, 2021.

2021 Lavinia Amorosi, Rossana Cavagnini, Veronica Dal Sasso, Martina Fischetti, Valentina Morandi, and **Alice Raffaele**. Women Just Wanna Have OR: Young Researchers Interview Expert Researchers. *SN Operations Research Forum*, volume 1, 2, 2021.

2020 Rossana Cavagnini, Veronica Dal Sasso, Valentina Morandi, and **Alice Raffaele**. How to Organize an OR Workshop: the AIROYoung Experience. *SN Operations Research Forum*, volume 4, 34, 2020.

In-conference proceedings

2021 Dario Ostuni, **Alice Raffaele**, Romeo Rizzi, and Matteo Zavatteri. Faster and Better Simple Temporal Problems. *Proceedings of the 35th AAAI Conference on Artificial Intelligence*, 2021.

Book chapters

2027 Veronica Dal Sasso, Serena Fugaro, Valentina Morandi, Lorenzo Peirano, and **Alice Raffaele**. Beyond the Experiment: A Decade of AIROYoung. In Alice Raffaele, Martina Doneda, Dario Palasgo, and Marta Leonina Tessitore, editors, *Optimization between Determinism and Uncertainty. Proceedings of the 10th AIROYoung Workshop and Ph.D. School, University of Padua, Italy, February 9-13, 2026. AIRO Springer Series*, Cham, 2027. Springer International Publishing. To appear.

2023 **Alice Raffaele**, Matteo Zavatteri, Fabio Bazzoli, Marco Gussago, and Romeo Rizzi. A Two-Phase Approach to Evaluate and Optimize an Interlibrary Loan Service: The Case Study of Provincia di Brescia. In Matteo Cosmi, Lorenzo Peirano, Alice Raffaele, and Marcella Samà, editors, *Operations Research and Data Science in Public Services*, pages 17–33, Cham, 2023. Springer International Publishing.

Editorials

2027 **Alice Raffaele**, Martina Doneda, Dario Palasgo, and Marta Leonina Tessitore. Preface. In *Optimization between Determinism and Uncertainty. Proceedings of the 10th AIROYoung Workshop and Ph.D. School, University of Padua, Italy, February 9-13, 2026. AIRO Springer Series*, Cham, 2027. Springer International Publishing. To appear.

2027 Maarten Löffler, **Alice Raffaele**, and Francesco Silvestri. Preface. In *Creative Mathematical Sciences Communication: 8th International Conference, CMSC 2026, Padua, Italy, October 12–15, 2026, Proceedings*, volume 16951 LNCS, pages v–vii, 2027.

Technical reports

2019 Jean-François Côté, Renata Mansini, and **Alice Raffaele**. *Tactical Time Window Management in Attended Home Delivery*. CIRRELT, 2019.

2019 Dusan Bikov, Raul Kangro, Harald Kitzmann, Petre Lameski, **Alice Raffaele**, and Zlatko Varbanov. Usage based insurance. European Study Group with Industry – ESGI 151, 2019.

Ph.D. theses

2022 **Alice Raffaele**. From optimization to listing: theoretical advances in some enumeration problems. Università degli Studi di Trento e Università degli Studi di Verona, 2022.

Books

2025 **Alice Raffaele**. *Teoremi della dualità in programmazione lineare*, volume 37 of *Rivoluzioni matematiche. I grandi Teoremi da Pitagora a Nash*. Le Scienze, October 2025.

Submitted works

2026 Gabriella Colajanni, Alessandro Gobbi, Marinella Picchi, **Alice Raffaele**, and Eugenia Taranto. Teaching Vehicle Routing Problems and Enhancing Problem-Solving with Competitive Learning. *Decision Sciences Journal of Innovative Education*, February 2026. November 2026 (major revision, under review).

2026 Antonino Carè, **Alice Raffaele**, Roberto Roberti, Sara Stoia, and Yanlu Zhao. What if Robots Could Take the Metro? Potential Benefits in Integrating Public Transport with Last-mile Delivery. *International Journal of Production Research*, March 2026. September 2026 (major revision, under review).

2025 Feilong Wang, Renata Mansini, **Alice Raffaele**, Filippo Ranza, and Roberto Roberti. A DP-based Exact Method for the Submodular Knapsack Problem. *INFORMS Journal on Computing*, November 2025. July 2026 (major revision, under review).

In progress

2027 **Alice Raffaele**, Roberto Roberti, and Feilong Wang. Synchronized Multimodal Urban Air Mobility. 2027. Provisional journal: Transportation Science. Expected submission date: March 31, 2027.

2027 **Alice Raffaele**, Carolina Osorio, and Roberto Roberti. Mathematical Programming for Discrete Simulation-based Optimization. 2027. Provisional journal: Transportation Science. Expected submission date: August 31, 2027.

2027 Tobias Huf, Roberto Roberti, **Alice Raffaele**, and Manuel Ostermeier. A Branch-and-Price Algorithm for Optimal Truck-and-Robot Optimization. 2027. Provisional journal: Transportation Science. Expected submission date: March 31, 2027.

2027 Gabriella Colajanni, Alessandro Gobbi, Marinella Picchi, **Alice Raffaele**, and Eugenia Taranto. Coopetitive learning for enhancing problem-solving skills through an optimisation problem-based challenge. 2027. Provisional journal: Mathematical Thinking and Learning. Expected submission date: February 28, 2027.

2027 Antonino Andrea Carè, Kaiky Maia, **Alice Raffaele**, Roberto Roberti, and Anand Subramanian. A new mathematical model and metaheuristic approaches for the two-machine scheduling problem with flexible periodic maintenance, job rejection, and weighted sum of completion times. 2027. Provisional journal: Journal of Heuristics. Expected submission date: February 28, 2027.

2026 **Alice Raffaele**, Demetrio Laganà, and Francesca Maggioni. Robust and Distributionally Vehicle Robust Routing Problems: A Comprehensive Review and Open Research Directions. 2026. Provisional journal: Computers & Operations Research. Expected submission date: December 31, 2026.

2026 Elisa Savio, **Alice Raffaele**, Laura Meneghetti, Pasquale Claudio Africa, and Gianluigi Rozza. Heuristic and proximal policy optimization approaches for a practical three-dimensional block relocation problem with item families. 2026. Provisional journal: Expert Systems with Applications. Expected submission date: November 30, 2026.

Magazines

2026 Veronica Dal Sasso and **Alice Raffaele**. Modeling a Community of Young Researchers: On the Origins of AIROYoung. *OR/MS Tomorrow*. INFORMS, June 2026. `https://www.informs.org/Publications/OR-MS-Tomorrow/ORTM_Spring_Summer_2026`.

2026 Maria Battarra, **Alice Raffaele**, and Roberto Roberti. EWG on Vehicle Routing and Logistics Optimization celebrated its 10th Meeting in Bath, UK. *IFORS News*, September 2026. `https://www.ifors.org/september-2026-issue/`.

2026 Manuel F. Aprile, Francesco Cavaliere, Martina Galeazzo, Anthony Palmieri, Dario Palasgo, Lorenzo Perinello, Matteo Pernini, **Alice Raffaele**, and Gabriele Sanguin. 10th AIROYoung Workshop and Ph.D. School, Celebrated in Padova, Italy - "Optimization Between Determinism and Uncertainty". *IFORS News*, June 2026. `https://www.ifors.org/june-2026-issue/`.

2025 **Alice Raffaele** and Anand Subramanian. Primal-Dual Partnerships. The interior of successful O.R. collaborations. *OR/MS Today*. INFORMS, September 2025. `https://pubsonline.informs.org/do/10.1287/orms.2025.03.06/full/`.

2025 Gabriella Colajanni, Alessandro Gobbi, Marinella Picchi, **Alice Raffaele**, and Eugenia Taranto. Ricerca operativa: attività di modellizzazione e problem-solving a scuola per minimizzare la distanza con la realtà. *Nuova Lettera Matematica*, volume 8, pages 86–100, 2025.

2024 **Alice Raffaele** and Anand Subramanian. If Dantzig Had a Sister. *OR/MS Today*. INFORMS, February 2024. `https://pubsonline.informs.org/do/10.1287/orms.2024.01.06/full`.

2024 **Alice Raffaele**. On Surnames, Symbols and Goals. EURO Gold Medal 2024 awarded to Maria Grazia Speranza. *OR/MS Today*. INFORMS, September 2024. `https://pubsonline.informs.org/do/10.1287/orms.2024.03.05/full/`.

2023 **Alice Raffaele** and Anand Subramanian. Revealing a Less-Traveled Road. *OR/MS Today*. INFORMS, September 2023. `https://pubsonline.informs.org/do/10.1287/orms.2023.03.03/full`.

2022 Marco Menale and **Alice Raffaele**. Recensione del libro "Uno, due, tre, molti, come la matematica ha creato la civiltà", di Michael Brooks. *Archimede: 4, 2022*, pages 223–224. Le Monnier, 2022.

2021 Lavinia Amorosi, Rossana Cavagnini, Veronica Dal Sasso, Martina Fischetti, Serena Fugaro, Valentina Morandi, and **Alice Raffaele**. Young OR enthusiasts at 4th AIROYoung Workshop in Bozen/Bolzano, Italy. *IFORS News*, March 2021. `https://ifors.org/newsletter/ifors-news-march2021.pdf`.

Talks

Invited

Dec, 2025 *"A combinatoria of experiences, information, books": Searching for my own optimal solution* at the *Sharing know-how, research lines and career tips: a symposium for early stage researchers*, ShaKER2025, December 04, 2025, Verona, Italy

Oct, 2025 *Emerging paradigms and new models in urban air mobility, last-mile delivery, and fleet sizing* at the *EURO WISDOM Forum YoungWomen4OR Talks, "OR for Routing and Transportation"*, October 02, 2025, online.

Jun, 2024 *Robust Policies for a Multi-Stage Assignment Problem under Demand Uncertainty* at the *GAMMA24* workshop, June 24, 2024, Verona, Italy.

Jun, 2023 *Exploring Innovative Approaches and Educational Opportunities to Decision Making*, third roundtable of the *1st DSA Summer International Conference*, with Alvaro Garcia Sanchez and Giovanni Righini, June 9, 2023, Milan, Italy.

Oct, 2022 *Introduction to graph theory: optimization problems and applications to social networks* for the *OPS4Math project – Optimization and Problem Solving for Teaching of Mathematics*, October 4, 2022, Naples, Italy.

Oct, 2022 *Solving optimization problem with GeoGebra* for the *OPS4Math project – Optimization and Problem Solving for Teaching of Mathematics*, October 4, 2022, Naples, Italy.

Nov, 2019 *A new decomposition for the Monotone Boolean Duality problem* at RIMS, Kyoto University, November 28, 2019, Kyoto, Japan.

Conferences

Sep, 2026 *Synchronized Multimodal Urban Air Mobility* at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2026, September 09, 2026, Galzignano Terme, Italy

Jul, 2026 *Synchronized Multimodal Urban Air Mobility* at the *EURO Working Group on Vehicle Routing and Logistics Optimization*, VeRoLog 2026, July 08, 2026, Bath, United Kingdom.

Sep, 2025 *The Multimodal Urban Air Mobility Problem with Synchronization Constraints* at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2025, September 03, 2025, Milan, Italy

Sep, 2025 *On communicating Operations Research: Raising awareness and becoming visible* at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2025, September 02, 2025, Milan, Italy.

Jun, 2025 *Last-mile Delivery with Autonomous Robots and Public Transport Lines* at the *EURO Working Group on Vehicle Routing and Logistics Optimization*, VeRoLog 2025, 16–19 June, 2025 Trento, Italy.

Sep, 2024 *Robust Policies for a Multi-Period Fleet Sizing Problem with Demand Uncertainty* at the *11th INFORMS Transportation Science and Logistics Society Workshop* TSL 2024, September 26, 2024, Nantes, France.

Jul, 2024 *Robust Policies for a Multi-Stage Assignment Problem under Demand Uncertainty* at the *EURO 2024* international conference, July 01, 2024, Copenhagen, Denmark.

Jul, 2024 *The ROAR project: How to improve Grade 10-12 interest and motivation in Mathematics through a learning path based on Operations Research* at the *EURO 2024* international conference, July 01, 2024, Copenhagen, Denmark.

May, 2024 *Robust Policies for a Multi-Stage Assignment Problem under Demand Uncertainty* at the *ODYSSEUS 2024* international conference, May 19–24, 2024, Carmona, Spain.

Feb, 2024 *Robust Policies for a Multi-Period Assignment Problem under Demand Uncertainty* at the *8th AIROYoung Workshop – Emerging technologies for decision support systems and innovative optimization paradigms* (8AYW), February 14–16, 2024, Rende, Italy.

Sep, 2023 *A two-phase approach to evaluate and optimize an interlibrary loan service: the case study of Provincia di Brescia* at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2023, September 04–07, 2023, Ischia, Italy.

Sep, 2023 *An Operation Research based Didactic Unit for Grade 12: The ROAR Experience, Part III* at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2023, September 04–07, 2023, Ischia, Italy.

Nov, 2022 *ROAR: How to improve Grade 10-12 interest and motivation in mathematics through a learning path based on Operations Research* at the *4th Scientix International Conference, November 18-19, 2022*, Online.

Sep, 2022 *On Vertex Enumeration and Hypergraph Dualization: relevance in combinatorial optimization and a new decomposition approach* at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2022, August 30 – September 02, 2022, Florence, Italy.

Feb, 2022 *The case study of Province of Brescia: analysis, evaluation, and improvement of an interlibrary loan service* at the *6th AIROYoung Workshop – Operations Research and Data Science in Public Services* (6AYW), February 23–25, 2022, Rome (hybrid), Italy.

Sep, 2021 *An Operation Research based Didactic Unit for Grade 10: The ROAR Experience, Part I* at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2021, September 14–17, 2021, Rome, Italy.

Feb, 2021 *Teaching Operations Research Before University: a focus on Grades 9–12* at the *5th AIROYoung Workshop – Optimization and Data Science: Trends and Applications* (4AYW), February 10–12, 2021, Naples (virtual), Italy.

Dec, 2020 *Faster listing of bonds based on biconnectivity* at the *4th International Workshop on Enumeration Problems & Applications*, WEPA 2020, December 6–10, 2020, Haifa (virtual), Israel.

Oct, 2019 *A new decomposition for the Monotone Boolean Duality problem* at the *3rd International Workshop on Enumeration Problems & Applications*, WEPA 2019, October 28–31, 2019, Awaji Island, Japan.

Sep, 2019 *Teaching Operations Research before University* at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2019, September 4–7, 2019, Genova, Italy.

Nov, 2018 *A Joint Generation approach for the Vertex Enumeration Problem* at the *2nd International Workshop on Enumeration Problems & Applications*, WEPA 2018, November 4–8, 2018, Pisa, Italy.

Sep, 2016 *The Stochastic Multi-period Time Windows Assignment Problem* at the *46th Annual Conference of the Italian Operations Research Society*, September 6–9, 2016, Trieste, Italy.

Posters

Feb, 2021 *Faster and Better Simple Temporal Problems*, at the *35th AAAI Conference on Artificial Intelligence*, February, 2–9, 2021, Online.

Industrial experience

Nov, 2016 – Oct, 2017 **IT manager**, *Edelweiss Energia S.p.A.*, Sarnico, Bergamo, Italy.
Apprenticeship in collaboration with University of Brescia and the Department of Information Engineering. Computer science engineer and system administrator, developer of some internal software, report manager, technical support to other areas of the company.

May, 2016 – Oct, 2016 **Internship**, *Edelweiss Energia S.p.A.*, Sarnico, Bergamo, Italy.
Implementation of ETL scripts to automatic download and load data from several sources; learning of the main company tools and software; technical support to other areas of the company.

Collaborations with companies

May, 2019 – Aug, 2023 **Consultant**, *Provincia di Brescia*, Brescia, Italy, Libraries Office.

Oct, 2018 – Mar, 2019 **Consultant**, *Consorzio Agrario del Nordest*, Verona, Italy.

Oct, 2017 – May, 2020 **Consultant**, *InfogestWeb s.r.l.*, Verona, Italy.
Joint Project 2018, granted by the University of Verona on the project entitled *DAVIDE (Development of Algorithms to Verify Infractions of Driving European laws)*. Principal Investigator: Prof. Romeo Rizzi.

Teaching at university level

Teaching and course responsibility

Oct, 2022 – Feb, 2024 **Mathematics for Decisions (contract professor)**, *University of Verona*, Italy.
Master's degree in *Mathematics* (English, 48 hours, classroom), average student evaluation: 3.55/5.

Teaching

Jan, 2023 – Feb, 2023 **Advances in Operations Research: Mathematical Programming (lecturer)**, *University of Trento*, Italy.
Ph.D. course in *Mathematics* (English, 12 hours, Zoom).

Mar, 2022 – Mar, 2023 **Discrete Optimization and Decision Making (contract professor)**, *University of Verona*, Italy.
Master's degree in *Data Science* (English, 32 hours, classroom + Zoom), average student evaluation: 3.41/5.

Mar, 2022 – Jun, 2022 **Operations Research (contract professor)**, *University of Verona*, Italy.
Bachelor's degree in *Applied Mathematics* (Italian, 8 hours, classroom + Zoom, average student evaluation: 3.75/5).

Oct, 2021 – Jan, 2022 **Mathematics for Decisions (contract professor)**, *University of Verona*, Italy.
Master's degree in *Mathematics* (English, 8 hours, classroom + Zoom, average student evaluation: 3.5/5).

Teaching assistant

Oct, 2020 – Jan, 2021 **Mathematics for Decisions**, *University of Verona*, Italy.
Master's degree in *Mathematics* (English, 24 hours, classroom + Zoom).

Mar, 2020 – Jun, 2020 **Operations Research**, *University of Verona*, Italy.
Bachelor's degree in *Applied Mathematics* (Italian, 24 hours, Zoom).

Mar, 2020 – Jun, 2020 **Mathematics for Decisions**, *University of Verona*, Italy.
Master's degree in *Mathematics* (English, 24 hours, Zoom).

Mar, 2019 – Jun, 2019 **Operations Research**, *University of Verona*, Italy.
Bachelor's degree in *Applied Mathematics* (Italian, 24 hours, classroom).

Mar, 2018 – Jun, 2018 **Operations Research**, *University of Verona*, Italy.
Bachelor's degree in *Applied Mathematics* (Italian, 24 hours, classroom).

Mini-courses and seminars

Nov, 2017 – Dec, 2017 **A hands-on tutorial on LP and ILP modeling (lecturer)**, *University of Verona*, Italy.
Mini-course in the Master's degree in *Mathematics* (20 hours).

Apr, 2016 **Adaptive Large Neighborhood Search (ALNS) (lecturer)**, *University of Brescia*, Italy.
Tutorial lecture in the *Optimization Algorithms* course by Prof. Renata Mansini (2 hours).

Memberships

Oct, 2021 – Mar, 2024 **Consiglio del Corso di Studio delle Lauree Triennali e Magistrali in Matematica**, *University of Verona*, Italy.
Department of Computer Science.

Advising – Ph.D. students

Jul, 2024 – Dec, 2026 **Elisa Savio (co-advisor)**, *SISSA*, Italy.
Industrial Ph.D. student (PNRR DM 352, in collaboration with Danieli Automation S.p.A.) in *Mathematical Analysis, Modelling, and Applications*.
Thesis: Operations Research and Reinforcement Learning Approaches for the Three-Dimensional Block Relocation Problem with Item Families.
Advisor: Prof. Gianluigi Rozza (SISSA). Other co-advisors: Dr. Pasquale Claudio Africa and Dr. Laura Meneghetti (SISSA).

Advising – Master's students

Jul, 2024 – Mar, 2025 **Antonino Andrea Carè (co-advisor)**, *University of Padova*, Italy.
Master's thesis in *Computer Engineering*.
Thesis: "MILP Models for Public Transport-aided Robot Routing".
Advisor: Prof. Roberto Roberti (University of Padova).
Note: since September 2025, Carè is a Ph.D. student at Politecnico di Milano.

Nov, 2023 – Oct, 2024 **Bernis Çolakoğlu (co-advisor)**, *University of Padova*, Italy.
Master's thesis in *ICT for internet and multimedia*.
Thesis: "Matheuristic Approaches for a Practical Flow Shop Scheduling Problem".
Advisor: Prof. Roberto Roberti (University of Padova).
Note: from December 2024 to October 2026, Çolakoğlu was a Ph.D. student at Politecnico di Milano.

Jul, 2019 – Mar, 2020 **Enrica Pini (co-advisor)**, *University of Verona*, Italy.
Master's thesis in *Mathematics*.
Thesis: "Optimization of the flow reconstruction in traffic networks".
Advisor: Prof. Romeo Rizzi.

Third mission and public engagement

Teaching experimentations addressed to young students

Dec, 2018 – present **Math4Kids (creator and trainer)**, *Biblioteca comunale Fausto Sabeo*, Chiari, Italy.
Presentation of mathematical applications to pupils in Grades 3–5 through engaging problem-solving activities and logic-game workshops, offered free of charge at a public library.
Websites: `https://opac.provincia.brescia.it/en/library/chiari/math-4-kids` `https://aliceraffaele.github.io/communication/math4kids/`

Nov, 2026 **Science 4 All (trainer)**, *University of Padova, Italy*.
Organization of "*Le Olimpiadi di Ottimizzazione*", activities and games carried out together with Prof. Roberto Roberti (University of Padova) to introduce the bases of operations research and algorithm development to Grades 5-7 classes. For more details, see `https://scuole.science4all.it/attivita/le-olimpiadi-di-ottimizzazione/`.

May, 2026 **I Giovedì dell'Orientamento (trainer)**, *Oratorio CG 2000*, Chiari, Italy.
Learning path composed of several meetings for secondary school students to explore STEM subjects through active learning, games, and collaborative activities. Funded by the Comune di Chiari and the Oratorio CG 2000. Main responsible for one meetings dedicated to activities and games to introduce the bases of operations research and algorithm development to Grades 6-8 students. For more details, see `https://aliceraffaele.github.io/images/cg2000-giovedì-orientamento.jpg`.

Nov, 2025 **Carnevale della Matematica dal Vivo 2025 (trainer)**, *Museo Nazionale Scienza e Tecnologia Leonardo da Vinci*, Milan, Italy.
Organization of "*Le Olimpiadi di Ottimizzazione*", activities and games carried out to introduce the bases of operations research and algorithm development to Grades 9 classes. For more details, see `https://maddmaths.simai.eu/carnevale-milano-leonardo-2025/`.

Nov, 2025 **Science 4 All (trainer)**, *University of Padova*, Italy.
Organization of "*Le Olimpiadi di Ottimizzazione*", activities and games carried out together with Prof. Roberto Roberti (University of Padova) to introduce the bases of operations research and algorithm development to Grades 5-7 classes. For more details, see `https://scuole.science4all.it/attivita/le-olimpiadi-di-ottimizzazione/`.

Nov, 2024 **Science 4 All (trainer)**, *University of Padova*, Italy.
Partecipation to "*Informatica senza computer*", activities and games carried out by Prof. Francesco Silvestri (University of Padova) to introduce the bases of informatics and algorithm development to two Grade 4 classes. For more details, see `http://scuole.science4all.it/attivita/informatica-senza-computer/`.

Jan, 2024 **La matematica del fare meglio (trainer)**, *Istituto Superiore "M.O. Luciano Dal Cero"*, San Bonifacio, Verona, Italy.
Participation in a three-hour educational module designed by Prof. Roberto Roberti and included in the University of Padova's PNRRorienta programme, aimed at upper secondary school students to introduce mathematical optimization as a tool for addressing real-world decision-making problems. For more details, see `https://pnrrorienta.unipd.it/`.

Nov, 2022 – Jan, 2023 **Banco di Prova (creator, designer, and experimenter)**, *Oratorio CG 2000*, Chiari, Italy.
Learning path designed with Dr. Davide Capoferri (Ph.D. in Biotechnology) and Dr. Giovanni Macetti (Ph.D. in Chemistry), composed of nine meetings for primary and secondary school students to explore STEM subjects through active learning, games, and collaborative activities. Funded by the Comune di Chiari and the Oratorio CG 2000. Main responsible for the three meetings dedicated to mathematics and informatics: the Caesar Cipher and other secret languages (Nov. 14, 2022); the Christmas (Knapsack) problem and an innovative bingo (Nov. 19, 2022); and the algorithm behind cooking pasta and sorting games with numbers and heights (Jan. 09, 2023).

Oct, 2022 – Jan, 2023 **ROAR III (creator, designer, and experimenter)**, *IIS Antonietti*, Iseo, Italy.
Path for Transversal Skills and Orientation in collaboration with Prof. Marinella Picchi, Dr. Alessandro Gobbi (University of Brescia), Dr. Eugenia Taranto and Dr. Gabriella Colajanni (University of Catania). Third part of a three-year project-work to introduce operations research to Grades 10–12. Focus on implementation and resolution of optimization problems by means of Python and the library PuLP. For further details, see `https://github.com/aliceraffaele/ROAR`.

Jan, 2022 – Apr, 2022 **ROAR II (creator, designer, and experimenter)**, *IIS Antonietti*, Iseo, Italy.
Path for Transversal Skills and Orientation in collaboration with Prof. Marinella Picchi, Dr. Alessandro Gobbi (University of Brescia), Dr. Eugenia Taranto and Dr. Gabriella Colajanni (University of Catania). Second part of a three-year project-work to introduce operations research to Grades 10–12. Focus on graph theory and network applications. For further details, see `https://github.com/aliceraffaele/ROAR`.

Mar, 2021 – May, 2021 **ROAR I (creator, designer, and experimenter)**, *IIS Antonietti*, Iseo, Italy.
Path for Transversal Skills and Orientation in collaboration with Prof. Marinella Picchi, Dr. Alessandro Gobbi (University of Brescia), Dr. Eugenia Taranto and Dr. Gabriella Colajanni (University of Catania). First part of a three-year project-work to introduce operations research to Grades 10–12. Focus on mathematical modelling, linear and integer programming. For further details, see `https://github.com/aliceraffaele/ROAR`.

Sep, 2019 **KidsUniversity (trainer)**, *University of Verona*, Italy.
Activities for classes of primary and secondary schools (8 hours).

Nov, 2018 – Dec, 2018 **KidsUniversity (trainer)**, *University of Verona*, Italy.
Problem solving in primary schools. Mathematics and informatics activities in thirteen classes in three primary schools in Bovolone and Dossobuono, Verona (26 hours).

Sep, 2018 **KidsUniversity (trainer)**, *University of Verona*, Italy.
Activities for classes of primary and secondary schools (14 hours).

Training courses for school teachers

Feb, 2023 – May, 2023 **ROAR Teacher Training Course (creator, designer, and lecturer)**, *ITSOS Marie Curie, Cernusco sul Naviglio*, Milan, Italy.
Organization of a teacher training course based on the first two teaching units of the ROAR project, aimed at introducing teachers to operations research and its applications in education. Delivered two sessions: "Introduction to graph theory: applications to social networks and network optimization problems" (February, 21, 2023) and "The origins of graph theory: the Königsberg bridge problem" (March, 6 2023).

Oct, 2022 **OPS4Math Project (guest lecturer)**, *Department of Electrical Engineering and Information Technology, University of Naples Federico II*, Naples, Italy.
Invited to deliver two seminars as part of the OPS4Math project, aimed at around 30 upper secondary school teachers in the Campania region, with the goal of promoting problem-solving skills and optimization algorithms in mathematics education. Delivered two sessions about solving two-variable optimization problems using GeoGebra and the introduction to graph theory through social networks and classical network optimization problems. For more details, see `https://ops4math.dieti.unina.it/index.php/it/`.

Science communication

Aug, 2021 – present **MaddMaths! (collaborator and writer)**, *online*, Italy.
Italy's leading platform for mathematics communication and outreach. I contribute columns and magazine articles on operations research, mathematics, scientific events, and women in STEM. Since January 2025, I have served as editor of "Storie che contano", a collection of short stories submitted by the platform's audience. MaddMaths! received an Honourable Mention in the 2026 European Mathematical Society Simon Norton Prize for Mathematics Outreach; I contributed to the preparation of the award application.
Websites: `https://maddmaths.simai.eu`
`https://euromathsoc.org/committee-outreach-Simon-Norton-Prize-2026`
`https://aliceraffaele.github.io/communication/maddmaths/`
`https://maddmaths.simai.eu/storie-che-contano/`

Mar, 2026 **Inspiring Girls Italia (role model)**, *I.C. 2e4 – Scuola Secondaria Bortolan*, Vicenza, Italy.
International project is to increase young students' awareness of their own talents, overcoming gender stereotypes that could hinder their careers. The project is carried out by hundreds of female volunteers from various professional fields who visit schools to discuss their personal and professional experiences.
Website: `https://www.inspiring-girls.it`.

Jun, 2025 **La rete e l'ottimizzazione (invited seminar)**, *Museo della Città*, Chiari, Italy.
1-hour seminar addressed to a general audience to provide an overview on some optimization problems on networks and to discuss the power of mathematical modeling and abstraction with respect to their infinite applications in the real world. The seminar was the last one of a series organized in parallel to the art exhibition "La rete" by Giorgio Tentolini, curated by Colossi Arte Contemporanea and held in Chiari, Italy, from May 24 to June 20, 2025. For further details, see `https://drive.google.com/file/d/1GlIaQjIjRINjJx9eBGZxN-pm5qugxocc/view` and `https://shorturl.at/Y58Kk`.

Sep, 2023 – May 2024 **Penne Amiche della Scienza (volunteer correspondent)**, *online*, Italy.
Continued participation in an Italian science outreach initiative inspired by Letters to a Pre-Scientist. Corresponding with a Grade-7 class at a middle school in Abruzzo. For more details, see `https://sites.google.com/view/penne-amiche-della-scienza`.

Dec, 2023 **I migliori libri di scienza scelti dai divulgatori scientifici (contributor)**, *online*, Italy.
A collaborative annual initiative by Giorgio Sestili bringing together science journalists, communicators, researchers, and scientists to recommend their favourite science books, essays, and science-related novels. For more details, see `https://www.giorgiosestili.it/migliori-libri-divulgazione-scienza-2023/`.

Oct, 2023 **Crescere con la scienza (host)**, *online*, Italy.
Online presentation of the catalog "Crescere con la scienza" (i.e., "Growing with science") by Editoriale Scienza together with the curator Angela Dal Gobbo and the director and founder of the publishing house Sabina Stavro. For more details, see `https://maddmaths.simai.eu/divulgazione/letture-matematiche/crescere-con-la-scienza-sperimentando-con-la-lettura/`.

Dec, 2022 **I migliori libri di scienza scelti dai divulgatori scientifici (contributor)**, *online*, Italy.
A collaborative annual initiative by Giorgio Sestili bringing together science journalists, communicators, researchers, and scientists to recommend their favourite science books, essays, and science-related novels. For more details, see `https://www.giorgiosestili.it/migliori-libri-divulgazione-scienza-2022/`.

Sep, 2022 – May, 2023 **Penne Amiche della Scienza (volunteer correspondent)**, *online*, Italy.
An Italian science outreach initiative inspired by Letters to a Pre-Scientist. Corresponded with a Grade-5 class at Istituto Comprensivo Giovanni Pascoli in Cento, Ferrara, during the 2022/2023 school year. For more details, see `https://sites.google.com/view/penne-amiche-della-scienza`.

Sep, 2022 – Apr, 2023 **Letters to a Pre-Scientist (volunteer correspondent)**, *online*, USA.
A science outreach initiative connecting STEM professionals with high school students through handwritten letters exchanged by snail mail, with the aim of broadening students' awareness of STEM careers, challenging stereotypes about STEM professionals, and inspiring them to pursue a future in STEM. For more details, see `https://prescientist.org/`.

Feb, 2022 **La scienza è per tutti... un sogno. Disegna una scienziata o uno scienziato (designer and referee)**, *Biblioteca Fausto Sabeo, Chiari*, Italy.
Second edition of the creative drawing competition for children aged 12 and under, organized in collaboration with Biblioteca Fausto Sabeo to celebrate the International Day of Women and Girls in Science (11 February). For more details, see `https://opac.provincia.brescia.it/library/chiari/concorso-creativo-la-scienza-e-per-tutti/`.

Nov, 2021 **Inspiring Girls Italia (role model)**, *I.C. Falcone-Borsellino*, Lazise, Verona, Italy.
International project is to increase young students' awareness of their own talents, overcoming gender stereotypes that could hinder their careers. The project is carried out by hundreds of female volunteers from various professional fields who visit schools to discuss their personal and professional experiences. Delivered two one-hour meetings in Grades 7-8 classes. Website: `https://www.inspiring-girls.it`.

Feb, 2021 **La scienza è per tutti... un sogno. Disegna una scienziata o uno scienziato (designer and referee)**, *Biblioteca Fausto Sabeo*, Chiari, Italy.
A creative drawing competition for children aged 12 and under, organized in collaboration with Biblioteca Fausto Sabeo to celebrate the International Day of Women and Girls in Science (11 February), encouraging young participants to imagine and portray women and men working in science. For more details, see `https://opac.provincia.brescia.it/library/chiari/concorso-creativo-la-scienza-e-per-tutti/`.

Service

Peer reviewing

2021 – present: AIRO Springer Series, Applied Mathematical Modelling, Discrete Applied Mathematics, European Journal of Operational Research, Heliyon, International Journal of Mathematical Education in Science and Technology, INFORMS Transactions on Education, International Transactions in Operational Research, Integer Programming and Combinatorial Optimization, Networks, Operations Research Forum, The Art of Discrete and Applied Mathematics, Transportation Science.

Book editing

2025 – present: **Special Volume editor** (with Martina Doneda, Dario Palasgo, and Marta L. Tessitore) of the book entitled *Optimization between determinism and uncertainty* in the AIRO Springer Series.

2022 – 2023: **Special Volume editor** (with Matteo Cosmi, Lorenzo Peirano, and Marcella Samà) of the book entitled *Operations Research and Data Science in Public Services* in the AIRO Springer Series.

Positions of responsibility

May, 2024 – present: **Representative** of AIRO in the MaddMaths! ETS association's first board.

Feb, 2021 – Feb, 2024: **Joint coordinator** of AIROYoung (Young Chapter of the Italian Operations Research Society).

Mar, 2019 – Feb, 2021: **Social media manager and web curator** of AIROYoung.

Organizing committees

Oct, 2026: *Creative Mathematical Sciences Communication 2026* (role: organizing **co-chair**), University of Padua, Italy, October 12-15, 2026, `http://cmsc2026.github.io`

Feb, 2026: *10th AIROYoung Workshop and Ph.D. School* (role: **chair**), University of Padua, Italy, February 9-13, 2026, `https://ayw2026.github.io`

Nov, 2025: *Carnevale della Matematica dal Vivo 2025* (role: **member**), Museo Nazionale Scienza e Tecnologia Leonardo da Vinci, Milan, Italy, November 21-22, 2025, `https://maddmaths.simai.eu/carnevale-milano-leonardo-2025/`

Feb, 2020: *4th AIROYoung Workshop* (role: **member**), Free University of Bozen, Italy, February 5–7, 2020

Program committees

2025: *International Conference on Optimization and Decision Science* (ODS2025), September 01–04, 2025, Milan, Italy.

2024: *International Conference on Optimization and Decision Science* (ODS2024), September 08–12, 2024, Badesi, Sardinia, Italy.

2024: *8th AIROYoung Workshop – Emerging technologies for decision support systems and innovative optimization paradigms* (8AYW), February 14–16, 2024, Rende, Italy.

2023: *7th AIROYoung Workshop – Operations Research Beyond Frontier* (7AYW), February 15–17, 2023, Milan, Italy.

2022: *International Conference on Optimization and Decision Science* (ODS2022), August 30 – September 02, 2022, Florence, Italy.

2022: *6th AIROYoung Workshop – Operations Research and Data Science in Public Services* (6AYW), February 23–25, 2022, Rome (hybrid), Italy.

Session chairs

Jul, 2027: *TBD* in the "Combinatorial Optimization" stream, at the *35th European Conference on Operational Research* (EURO2027), July 11–14, 2027, Athens, Greece.

Sep, 2025 *Communicating mathematics and O.R. (MaddMaths!)* at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2025, September 02, 2025, Milan, Italy.

Jul, 2024 *Vehicle Routing Under Uncertainty 1* in the "VeRoLog – Vehicle Routing and Logistics" stream, *33rd European Conference on Operational Research* (EURO2024), July 1, 2024, Copenhagen, Denmark.

Jul, 2024 *Experienced Routes for the Teaching Students' Problem* in the "OR Education" stream, *33rd European Conference on Operational Research* (EURO2024), July 1, 2024, Copenhagen, Denmark.

Sep, 2023 *AIROYoung Dissertation Award 2023* (with Veronica Dal Sasso, Giusy Macrina, and Lorenzo Peirano) at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2023, September 07, 2023, Ischia, Italy.

Sep, 2023 *Teaching and Education* (with Claudio Sterle) at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2023, September 05, 2023, Ischia, Italy.

Sep, 2022 *Best AIROYoung Dissertation 2022* at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2022, September 02, 2022, Florence, Italy.

Sep, 2022 *Discrete Optimization III* at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2022, September 01, 2022, Florence, Italy.

Sep, 2021 *OR IN... FIVE MINUTES – A pitch-talk session* (with Lorenzo Peirano) at the *International Conference on Optimization and Decision Science*, AIRO–ODS 2021, September 15, 2021, Rome, Italy.

Memberships

Dec, 2024 – present **VeRoLog** (EURO Working Group on Vehicle Routing and Logistics Optimization).

Apr, 2022 – present **OPSTeaM** (Mathematics Teaching Chapter of the Italian Operations Research Society).

Jan, 2021 – present **GNCS-INdAM** (National Group for Numerical Analysis – Gruppo Nazionale per il Calcolo Scientifico dell'Istituto Nazionale di Alta Matematica, Sezione Fondamenti di Informatica e Sistemi Informatici).

Dec, 2020 – Nov, 2021 **AAAI** (Association of the Advancement of Artificial Intelligence).

Mar, 2019 – present **AIROYoung** (Young Chapter of the Italian Operations Research Society).

Sep, 2016 – present **AIRO** (Italian Operations Research Society).

Achievements

Awards and honors

Oct, 2026 Recipient of the **Transportation Science Meritorious Service Award**, in appreciation of an exceptional service as a reviewer for *Transportation Science* (INFORMS).

Dec, 2024 Selected as **one of the 12 YoungWomen4OR 2024**, an initiative by the EURO WISDOM Forum. For more details, see `https://www.euro-online.org/web/pages/1654/wisdom`.

Jul, 2022 Selected as **one of the 104 participants** (among 730+ applications) of the first edition of **EU TalentOn**, an initiative by European Commission addressed to early career researchers between 21 and 35 years old.

Feb, 2022 Awarded of the **Best Paper Award** for the book chapter "*A Two-Phase Approach to Evaluate and Optimize an Interlibrary Loan Service: The Case Study of Provincia di Brescia*".

May, 2021 Selected as a **Reading Ambassador**, an Italian award issued by the Centro per il Libro e la Lettura to people and organizations that promote reading and foster best practices related to books and reading.

Sep, 2016 Awarded of the ***AIRO Prize*** for the best master thesis related to applications during the 46th Annual Conference of the Italian Operations Research Society in Trieste, Italy, September 6–9, 2016.

May, 2005 Awarded of the ***Journalist for a day*** prize in a national writing contest organized by the National Circuit of School Journalism – AlboScuole in Rome, Italy, May 20, 2005.

Grants

Nov, 2025 Recipient of a **sponsorship** by **Gruppo Nazionale per il Calcolo Scientifico (GNCS-INdAM)** to support the organization of the *Creative Mathematical Sciences Communication 2026* in October 2026 at the University of Padova. 800 €.

Jun, 2023 Recipient of a **conference grant** funded by **Gruppo Nazionale per il Calcolo Scientifico (GNCS-INdAM)** to attend the *International Conference on Optimization and Decision Science*, AIRO–ODS 2023, September 04–07, 2023, Ischia, Italy. 300 €.

Sep, 2022 Recipient of a **travel grant** by the **European Commission** to participate to the first edition of **EU TalentOn** in Leiden, The Netherlands, September 14–18, 2022.

Jan, 2022 Recipient of a **conference grant** funded by **Gruppo Nazionale per il Calcolo Scientifico (GNCS-INdAM)** to attend the *International Conference on Optimization and Decision Science*, AIRO–ODS 2022, August 30 – September 02, 2022, Florence, Italy. 500 €.

Feb, 2019 Recipient of **Cost Action TD1409 Travel Grants** to attend the ***First Estonian Study Group with Industry (ESGI 151)***, University of Tartu, Estonia, February 4–8, 2019. 384.98 €.

Sep, 2018 Recipient of **COST Action TD1409 Travel Grants** to attend the ***Modeling Week on Data Driven Decision Making and Optimization (D3EMO)***, Copenhagen Business School, Denmark, September 3–7, 2018. 400 €.

Nov, 2017 – Mar, 2021 Recipient of the **Italian PhD Scholarship** to attend the PhD course in Mathematics jointly organized by University of Trento and University of Verona. 16350 €/year.

Sep, 2015 – Jan, 2016 Recipient of the **Scholarship of Excellence** funded by Université Laval, Québec, Canada. 4000 CAD.

Sep, 2015 – Jan, 2016 Recipient of the **Thesis Abroad Scholarship** funded by University of Brescia to develop my master's thesis at Université Laval, Québec, Canada. 1500 €.

Competences and skills

Courses, schools, seminars, and workshops attended

Jul, 2026 *2nd VeRoLog PhD School*, July 03–06, 2026 (15 hours), Bath, United Kingdom.

Feb, 2026 *10th AIROYoung Workshop – Optimization between Determinism and Uncertainty* (10AYW), February 09–13, 2026, Padova, Italy.

Jun, 2025 *1st VeRoLog PhD School*, June 13–16, 2025 (24 hours), Trento, Italy.

Feb, 2025 *9th AIROYoung Workshop – Shaping a Sustainable Future in the Era of Big Data* (9AYW), February 26–28, 2025, Pavia, Italy.

Sep, 2024 *PhD School: Optimizing Transportation and Logistics under Uncertainty*. September 22–24, 2024, Nantes, France.

Jun, 2024 *GAMMA24 – General Advances in Mathematics and Models for Applications*, June, 24, 2024, Verona, Italy.

Feb, 2024 – Mar, 2024 *061652 Robust Optimization*. Lecturer: Prof. Erick Delage (HEC Montréal). February, 27 – March, 08, 2024 (27 hours), Milan, Italy.

Feb, 2024 *8th AIROYoung Workshop – Emerging technologies for decision support systems and innovative optimization paradigms* (8AYW), February 14–16, 2024, Rende, Italy.

Feb, 2023 *7th AIROYoung Workshop – Operations Research Beyond Frontier* (7AYW), February 15–17, 2023, Milan, Italy.

Feb, 2022 *6th AIROYoung Workshop: Operations Research and Data Science in Public Services* (6AYW), February 23–23, 2022, Rome (hybrid), Italy.

Feb, 2021 *AIRO PhD School 2021 & 5th AIROYoung Workshop – Optimization and Data Science: Trends and Applications* (5AYW), February 8–12, 2021, Naples (virtual), Italy.

Feb, 2020 *4th AIROYoung Workshop: New Advances in Optimization, Machine Learning and Data Science* (4AYW), February 5–7, 2020, Bozen, Italy.

Dec, 2020 *2nd Workshop On Graphs and Geometric Algorithms* (WOGGA2), National Institute of Informatics, Tokyo, Japan, December 16–18, 2019 (20 hours).

Mar, 2019 *3rd AIROyoung Workshop and 1st AIROYoung PhD School: Advanced Methods in Optimization and Data Science* (3AYW), Sapienza University of Rome, Department of Statistics, Rome, Italy, March 26–29 (24 hours).

Mar, 2019 *2nd conference of the EURO Working Group on the Practice of Operations Research*, Bologna, Italy, March 11–12, 2019 (16 hours).

Feb, 2019 *First Estonian Study Group with Industry* (ESGI 151), University of Tartu, Estonia, February 4–8, 2019 (40 hours).

Sep, 2018 *Modeling Week on Data Driven Decision Making and Optimization* (D3EMO), Copenhagen Business School, Denmark, September 03–07, 2018 (40 hours).

Jul, 2018 *Metodologie di osservazione etnografica in aula e nella scuola*. Lecturer: Prof. Elisa Bellato (University of Verona). Verona, July 24, 30–31, 2018 (18 hours).

Jul, 2018 *Elementi di didattica generale*. Lecturer: Prof. Giuseppina Messetti (University of Verona). Verona, July 20–21, 23, 26–28, 2018 (36 hours).

Jul, 2018 *Teorie della scuola e della formazione*. Lecturer: Prof. Daniele Loro (University of Verona). Verona, July 16–18, 2018 (18 hours).

Jul, 2018 *Pedagogia e didattica dell'inclusione* Lecturer: Prof. Angelo Lascioli (University of Verona). Verona, July 11–13, 2018 (18 hours).

Jul, 2018 *Psicologia sociale*. Lecturer: Prof. Sabrina Berlanda (University of Verona). Verona, July 10, July 19 and August 01, 2018 (18 hours)

Jun, 2018 – Jul, 2018 *Psicologia dell'apprendimento scolastico*. Lecturer: Prof. Laura Fontecedro (University of Verona). Verona, June 28–29 and July 02, 2018 (18 hours).

Jun, 2018 *Storia e attuale ordinamento delle istituzioni scolastiche*. Lecturer: Prof. Mario Gecchele (University of Verona). Verona, June 25–27, 2018 (18 hours).

Jun, 2018 *The Applications of Linear Programming*, University of Trento, Italy, June 18–19, 2018 (8 hours).

Jun, 2018 *School on Graph Theory* (SGT 2018), LIRMM, Sète, France, June 11–15, 2018 (36 hours).

May, 2018 *International Spring School on Integrated Operational Problems* (SSIOP), Université de Troyes, France, May 14–16, 2018 (20 hours).

Apr, 2018 *Colouring, Flows and Decompositions of Graphs*. Lecturer: Edita Macajova (Comenius University, Bratislava, Slovacchia), University of Verona, Italy, April 17–26 2018 (12 hours).

Apr, 2018 – May, 2018 *Introduction to Quantum Information*. Lecturer: Dr. Davide Pastorello. Trento, April - May 2018 (20 hours).

Mar, 2018 *Neural Networks for Time Series Analysis*. Lecturer: Oleksandr Honchar (University of Verona), University of Verona, Italy, March 21–29, 2018 (8 hours).

Feb, 2018 – May, 2018 *Stochastic Analysis: From Population Dynamics to Finance*. Lecturers: Prof. Luca Di Persio (University of Verona), Prof. Simone Scotti (University Paris Diderot, France) and Prof. Zenghu Liu (Beijing Normal University, China), Verona, Italy, February – May 2018 (60 hours).

Jan, 2018 – Feb, 2018 *Mathematical Programming*. Lecturer: Prof. Romeo Rizzi. Trento, January 22 – February 02, 2018 (20 hours).

Jan, 2018 *Introduction to SageMath*. Lecturer: prof. Péter Burcsi (Faculty of Informatics, Eötvös Loránd University, Budapest), University of Verona, Italy, January 22–25, 2018 (8 hours).

Languages

Italian Mother tongue

English Professional proficiency

French Elementary proficiency

Computer skills

Programming Languages C, C++, Python, Java, SQL, C#, HTML, Markdown, CSS, PHP, Javascript

Technologies AMPL, AndroidStudio, Eclipse, Gurobi, LaTeX, MATLAB, Microsoft Office (Word, Excel, Outlook, Powerpoint, Publisher), Microsoft SQL Server, Microsoft Visual Studio, Pentaho Kettle, VMware vSphere, SageMath, SAP Business Objects – Desktop Intelligence, Webratio Platform

Certifications

Apr, 2017 EPEX Spot Certification

Jul, 2016 State Professional Qualifying Examination in Engineering (Esame di Stato per l'abilitazione alla professione di Ingegnere – Sez. A, Ingegneria dell'Informazione)

May, 2009 PET B1 Preliminary

May, 2008 ECDL license

Volunteering

Jan, 2017 – Feb, 2024 **"I MiseraLibri" bookclub (founder and coordinator)**, *Biblioteca comunale Fausto Sabeo*, Chiari, Italy.
Facebook: `https://www.facebook.com/groups/GruppoDiLetturaChiari`
Google Maps:
`https://www.google.com/maps/d/u/0/edit?mid=1DofkuuHNFXbRL1pQStHPt37R76AAPcY&usp=sharing`

Jan, 2021 – Jul, 2021 **4th National Meeting of Italian Bookclubs (creator and member of the organizing committee)**, Chiari, Italy.
Organizer of four Zoom meetings to discuss several typologies and aspects of Italian bookclubs and chair of two of them. Organizer and chair of the 4th National Meeting of Italian Bookclubs, held on June 26th, 2021, at Villa Mazzotti in Chiari (Brescia, Italy).

Dichiarazione Sostitutiva di Certificazione e dell'Atto di Notorietà
La sottoscritta Alice Raffaele, nata a Chiari (BS) il 17 ottobre 1991, residente a Chiari (BS) in Via Sala 52, ai sensi degli artt. 46 e 47 del D.P.R. 445/2000 e consapevole delle sanzioni penali previste dagli artt. 75 e 76 del medesimo e s.m.i. per ipotesi di falsità in atti e dichiarazioni mendaci, DICHIARA, sotto la propria personale responsabilità, che quanto riportato nel presente curriculum vitae corrisponde a verità.

Vicenza, 09 novembre 2026

Autorizzo il trattamento dei miei dati personali ai sensi del Decreto Legislativo 30 giugno 2003, n. 196 e del GDPR (Regolamento UE 2016/679).

Vicenza, 09 novembre 2026